MuGyum Kim

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kmmugyum.github.io/portfolio · github.com/kmmugyum · linkedin.com/in/mugyum-kim

AI/ML undergraduate researcher in computer vision, generative models, and anomaly detection.

Education

B.Eng. in Artificial Intelligence

Chungnam National University, Daejeon, Republic of Korea

Mar. 2022 – Feb. 2028

Major GPA: 4.41/4.5 | Cumulative GPA: 4.23/4.5

Summer Program — AI and Generative Models (Certificate of Recognition)

University of Illinois Urbana-Champaign, Grainger College of Engineering, IL, USA

Jul. 2026 – Aug. 2026

Publications

MuGyum Kim*, GiWon Kim, ChanJong Park, SungHo Lee†, "Aggregation Sensitivity in Diffusion Inpainting Along a Mean–Median Axis," IEEE/IEIE ICCE-Asia 2026, pp. 396–401 (Oral). *Presenting author.

Skills

- **Languages & Frameworks:** Python, C/C++, PyTorch, TensorFlow, LoRA/PEFT
- **Libraries & Tools:** NumPy, Pandas, scikit-learn, OpenCV, Albumentations, Weights & Biases

Experience

Undergraduate Researcher

DIPLAB, Chungnam National University · Funded by Saemmulter, Inc.

Sep. 2026 – Oct. 2026

- Tested TCN and DLSTM groundwater forecasts against persistence on 58 wells with a single test opening.
- Found no significant model difference after Holm correction; delivered the report and an HTML dashboard.

Undergraduate Researcher

DIPLAB, Chungnam National University · Funded by ETRI

Mar. 2026 – Sep. 2026

- Trained a multi-label MLP on 28-dim SDS epoch features: 94.8% test macro-F1 on source-IP-disjoint splits.
- Ran inference on 16.2M unlabeled production epochs; deployed scanning detection as a ClickHouse service.

Undergraduate Researcher

DIPLAB, Chungnam National University, Daejeon, Republic of Korea

May 2023 – Feb. 2024

- Curated and labeled an 81,000+ sample satellite imagery dataset for cropland and greenhouse segmentation.
- Trained a SegFormer-based model reaching 88.9% mIoU on three-class cropland segmentation.
- Labeled a CCTV vehicle pose estimation dataset for a drunk-driving analysis pipeline.

Honors & Activities

- **Scholarships**: Academic Excellence (2022/2, 2023/1, 2023/2); SW-Centered University Activity (2023/2).
- **Undergraduate Tutor**, "AI and Future Society," CNU (2026): attendance, Q&A, and proctoring support.
- **Treasurer**, DIVE (AI Club), CNU (2023): organized a corn seed classification competition.